

The BCC Watson Reflection Bridge and the Master Quadratic

Foundational Ternary Dynamics Project

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Abstract

We isolate an exact arithmetic bridge between the body-centered cubic (BCC) Watson integral and the lemniscatic reflection ratio

$$G^* = \frac{\Gamma(1/4)}{\Gamma(3/4)}.$$

For the normalized BCC Watson integral,

$$W_{\text{BCC}} = \frac{1}{(2\pi)^3} \int_{[-\pi, \pi]^3} \frac{d^3 k}{1 - \cos k_x \cos k_y \cos k_z},$$

Watson's evaluation and Euler's reflection formula give

$$W_{\text{BCC}} = \frac{\Gamma(1/4)^4}{4\pi^3} = \frac{G^{*2}}{2\pi}.$$

After selecting the normalization $K := |\text{Aut}_{\mathbb{C}}(E, 0)|^2 (2\pi) W_{\text{BCC}} = 16G^{*2}$ for $E : y^2 = x^3 - x$, and after imposing the filtered product rule $P := G^* K$, the Vieta data are

$$x_+ + x_- = 16G^{*2}, \quad x_+ x_- = 16G^{*3},$$

and hence the master quadratic

$$x^2 - 16G^{*2}x + 16G^{*3} = 0.$$

Its larger root is $x_+ = 137.036171458\dots$, within 1.26 ppm of the measured inverse fine-structure constant. The equality with physical α^{-1} is not assumed as a theorem here; it is recorded as a phenomenological selection requiring independent physical matching.

1 The BCC Watson Integral

We consider the body-diagonal nearest-neighbor walk with steps $(\pm 1, \pm 1, \pm 1)$ on $\mathbb{Z}^3$, restricted to a parity class. Geometrically this is the eight-corner shell of the 26-connected Moore neighborhood, or the Stella Octangula corner shell; after the usual coordinate scaling it is the BCC nearest-neighbor pathing structure. The corresponding dimensionless Green denominator is

$$1 - \cos k_x \cos k_y \cos k_z.$$

Definition 1 (Normalized BCC Watson integral). *Define*

$$W_{\text{BCC}} = \frac{1}{(2\pi)^3} \int_{-\pi}^{\pi} \int_{-\pi}^{\pi} \int_{-\pi}^{\pi} \frac{dk_x dk_y dk_z}{1 - \cos k_x \cos k_y \cos k_z}.$$

By evenness and periodicity, this equals Watson's first triple integral

$$\frac{1}{\pi^3} \int_0^{\pi} \int_0^{\pi} \int_0^{\pi} \frac{du dv dw}{1 - \cos u \cos v \cos w}.$$

Theorem 1 (Watson reflection bridge). *Let*

$$G^* = \frac{\Gamma(1/4)}{\Gamma(3/4)}.$$

Then

$$W_{\text{BCC}} = \frac{\Gamma(1/4)^4}{4\pi^3} = \frac{G^{*2}}{2\pi}.$$

Proof. The evaluation

$$W_{\text{BCC}} = \frac{\Gamma(1/4)^4}{4\pi^3}$$

is Watson's first triple integral [1, 7]; see also the extended Watson-integral literature [2, 3]. Euler's reflection formula gives

$$\Gamma(1/4)\Gamma(3/4) = \pi\sqrt{2}.$$

This is the $z = 1/4$ case of $\Gamma(z)\Gamma(1-z) = \pi/\sin(\pi z)$ [4]. Therefore

$$G^* = \frac{\Gamma(1/4)}{\Gamma(3/4)} = \frac{\Gamma(1/4)^2}{\pi\sqrt{2}},$$

and so

$$\frac{G^{*2}}{2\pi} = \frac{\Gamma(1/4)^4}{4\pi^3} = W_{\text{BCC}}.$$

□

Theorem 1 is classical special-function and lattice-integral mathematics. Its interpretation as an FTD substrate identity is a model selection, not a consequence of the formula alone.

2 Capacity Normalization

The master quadratic uses the total normalized capacity of a selected BCC/CM sector. The arithmetic normalization is supplied by the CM elliptic curve

$$E : y^2 = x^3 - x,$$

which has $j(E) = 1728$ and complex automorphism group fixing the origin of order 4 [5]. Thus

$$|\text{Aut}_{\mathbb{C}}(E, 0)|^2 = 16.$$

Definition 2 (FTD capacity normalization). *For the selected BCC/CM sector, define the total capacity*

$$K = 16 \cdot (2\pi) \cdot W_{\text{BCC}}.$$

This is a normalization rule in the FTD construction. It is not derived from the Watson integral alone. Substituting Theorem 1 gives

$$K = 16 \cdot (2\pi) \cdot \frac{G^{*2}}{2\pi} = 16G^{*2}.$$

Numerically,

$$K = 140.060135374 \dots$$

3 The Master Quadratic

Definition 3 (Filtered master polynomial). *Given the capacity K above, define the filtered master polynomial as the monic quadratic whose elementary symmetric polynomials are*

$$e_1 = K, \quad e_2 = G^* K.$$

Equivalently, if x_+ and x_- are its roots, then

$$x_+ + x_- = K, \quad x_+ x_- = G^* K.$$

This definition encodes the self-reference rule of the FTD construction: the product term is one reflection-ratio step deeper than the trace term.

Corollary 1 (Master quadratic). *Under Definitions 2 and 3, the filtered master polynomial is*

$$x^2 - 16G^{*2}x + 16G^{*3} = 0.$$

Proof. By Vieta's formulas, the monic quadratic with elementary symmetric data $e_1 = K$ and $e_2 = G^* K$ is

$$x^2 - Kx + G^* K = 0.$$

Using Definition 2 and Theorem 1,

$$K = 16G^{*2}, \quad G^* K = 16G^{*3}.$$

Substitution gives the stated polynomial. □

The Watson/reflection identity is theorem-level. The capacity normalization and filtered product rule are definitions of the FTD master-quadratic construction. This separation is essential: the paper proves the algebraic consequences of the definitions; it does not prove that nature must choose them.

4 Roots and Phenomenological Remarks

The exact roots of the master quadratic are

$$x_{\pm} = \frac{16G^{*2} \pm \sqrt{(16G^{*2})^2 - 64G^{*3}}}{2}.$$

Equivalently,

$$x_{\pm} = 8G^{*2} \pm 4G^{*3/2} \sqrt{4G^* - 1}.$$

Evaluating with $G^* = 2.958675119188638 \dots$ gives

$$x_+ = 137.036171458155 \dots, \quad x_- = 3.023963916339 \dots$$

The CODATA 2022 low-energy inverse fine-structure constant is

$$\alpha(0)^{-1} = 137.035999177(21) \quad [6].$$

Thus x_+ differs from $\alpha(0)^{-1}$ by

$$1.72281 \times 10^{-4},$$

approximately 1.257 ppm. This is far outside the CODATA experimental uncertainty; any physical identification therefore requires additional FTD matching or correction terms. The mathematical result is exact; the physical identification

$$x_+ \stackrel{?}{=} \alpha(0)^{-1}$$

is a phenomenological selection requiring a separate matching from FTD lattice variables to physical electromagnetic normalization.

The smaller root lies near the integer 3. In the FTD interpretation, this motivates the conjectural association with a threefold confined sector, suggestive of the color rank $N_c = 3$. As with the electromagnetic identification, this is not a theorem of the quadratic alone: the rigorous statement is the exact value of x_- and its proximity to 3.

5 Epistemic Status

The results above separate into four layers:

1. The BCC Watson/reflection identity $W_{\text{BCC}} = G^{*2}/(2\pi)$ is an exact theorem.
2. The normalization $K = 16 \cdot 2\pi W_{\text{BCC}}$ and filtered product G^*K are explicit FTD definitions.
3. Given these definitions, the master quadratic follows by Vieta's formulas.
4. The identifications $x_+ \approx \alpha(0)^{-1}$ and $x_- \approx N_c$ are phenomenological selections, not established physical derivations in this paper.

6 Conclusion

We have isolated the exact arithmetic chain

$$W_{\text{BCC}} = \frac{G^{*2}}{2\pi} \implies K = 16 \cdot 2\pi W_{\text{BCC}} = 16G^{*2} \implies x^2 - 16G^{*2}x + 16G^{*3} = 0,$$

under the stated CM/BCC normalization and filtered self-reference rule. The larger root is

$$x_+ = 137.036171458\dots,$$

which is strikingly close to the low-energy physical inverse fine-structure constant, but not within experimental uncertainty. The paper therefore establishes a rigorous mathematical source for the master quadratic and records the physical interpretation as a sharply defined open matching problem rather than as an unconditional derivation of Standard Model parameters.

References

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